

1.1 1. Show that $|x^2 - 4| < \varepsilon$ when $0 < |x - 2| < \varepsilon(5 + \varepsilon)^{-1}$ and prove $\lim_{x \rightarrow 2} x^2 = 4$ by using these inequalities.

$$\boxed{1.1}$$

$$\boxed{1.}$$

given

$$\boxed{|x-2| < \frac{\varepsilon}{5+\varepsilon}} \quad (a)$$

We know:

$$|x+2| = |(x-2) + 4|$$

$$\text{And } |x+2| \leq |x-2| + 4$$

$$\text{or } |x+2| \leq \frac{\varepsilon}{5+\varepsilon} + 4$$

$$\text{or } |x+2| \leq \frac{20+5\varepsilon}{5+\varepsilon}$$

$$\text{or } \boxed{|x+2| \leq \frac{5(4+\varepsilon)}{(5+\varepsilon)}} \quad (b)$$

(a) $\times$ (b):

$$|x^2 - 4| \leq \underbrace{\frac{5}{5+\varepsilon}}_{< 1} \cdot \underbrace{\frac{4+\varepsilon}{5+\varepsilon}}_{< 1} \cdot \varepsilon$$

$$\Rightarrow |x^2 - 4| < \varepsilon$$

$\therefore \delta = \frac{\varepsilon}{5+\varepsilon}$ exists such that

$|n^2 - 4| < \varepsilon$ is true for
any $\varepsilon > 0$, where $0 < |n-2| < \delta$

$\therefore$ we have proved that

$$\lim_{n \rightarrow 2} n^2 = 4 \quad \square.$$



2. Show that the function $f(x) = x \sin(1/x)$, with $f(0) = 0$, is continuous at 0 but not differentiable at 0.

TPT: $f(x)$ is C at $x=0$.

Given $f(0) = 0$.

Left hand limit: $0-h = -h$
(LHL)

$$f(-h) = (-h) \sin\left(\frac{1}{-h}\right) = \underbrace{h}_{\sim 0} \underbrace{\sin\left(\frac{1}{h}\right)}_{\in (-1,1)} \rightarrow 0$$

$$\boxed{\text{LHL} = f(0^-) = 0}$$

Right hand limit:
(RHL): $0+h = h$

$$f(h) = h \sin\left(\frac{1}{h}\right) = \underbrace{h}_{\sim 0} \underbrace{\sin\left(\frac{1}{h}\right)}_{\in (-1,1)} \rightarrow 0$$

$$\boxed{\text{RHL} = f(0^+) = 0}$$

$$\therefore f(0^-) = f(0^+) = f(0) = 0$$

$$\therefore \text{Limit of } f(x) \text{ exists at } x=0.$$



check if $f(x)$ is D @ $x=0$

or check if $f'(x)$ is C @ $x=0$

$$\text{LHL of } f'(x): \frac{f(x) - f(x-h)}{h}$$

$x=0$:

$$f'(0^-) = \frac{f(0) - f(-h)}{h}$$

$$\text{or } f'(0^-) = \frac{0 - \left(h \sin \frac{1}{h}\right)}{h}$$

$$\text{or } \boxed{\begin{array}{l} f'(0^-) = -\sin\left(\frac{1}{h}\right) \\ \text{LHL} \end{array}}$$

$$\text{RHL of } f'(x) = \frac{f(x+h) - f(x)}{h}$$

$$f'(0^+) = \frac{f(h) - f(0)}{h}$$

$$\therefore f'(0^+) = \frac{h \sin \frac{1}{h} - 0}{h}$$

$$\therefore \boxed{f'_{\text{RHL}}(0^+) = \sin\left(\frac{1}{h}\right)}$$

$$\therefore f'(0^-) \neq f'(0^+) \Rightarrow \text{Limit of } f'(x) \text{ DNE @ } x=0.$$

$-\sin\left(\frac{1}{h}\right) \qquad \sin\left(\frac{1}{h}\right)$

$$\therefore f(x) \text{ is not D @ } x=0.$$

□

23. For small values of x , the approximation $\sin x \approx x$ is often used. Estimate the error in using this formula with the aid of Taylor's Theorem. For what range of values of x will this approximation give results correct to six decimal places?

$$\sin(x) = x + E_2(x)$$

$n=2$

Note: Technically

$$\sin(x) = x + E_1(x)$$

$n=1$

but since there is the x^2 term variables in the Taylor's series for $\sin(x)$, $E_1(x)$ is unnecessarily conservative (larger error). $E_2(x)$ can provide a smaller (yet correct) upper bound on the error.

$$E_2(x) = \frac{1}{3!} f^{(3)}(\xi) \cdot x^3$$

$$\text{or } E_2(x) = \frac{1}{6} x^3 \cdot \underbrace{(-\cos(\xi))}_{\in [-1, 1]}$$

$$|E_2(x)| \leq \frac{x^3}{6}$$

$$\text{or } |E_2(x)|_{\max} = \frac{x^3}{6}$$

$$\text{If } |E_2(n)|_{\max} < 1e-6$$

$$\bullet \frac{|n^3|}{6} < 1e-6$$

$$\bullet |n^3| < 6e-6$$

$$\bullet |n| < 6^{1/3} e^{-2}$$

$$\approx 1.04^\circ$$

$$\bullet \boxed{|n| < 1.8171e-2} \quad \underline{\text{Ans}}$$

Note, the looser $|E_1(n)|$ bound
 would have provided $\frac{1}{2} n^2 (\text{Max.})$
 $|n| < 1.414e-3$, which is
 much more constraining
 (unnecessary). $\approx 0.08^\circ$



31. Determine the Taylor polynomial of degree 2 for $f(x) = e^{(\cos x)}$ expanded about the point π .

$$f(x) \approx \sum_{k=0}^2 \frac{1}{k!} \cdot f^{(k)}(c) \cdot (x-c)^k$$

$x=2$

here $c = \pi$.

$$f(c) = e^{\cos c} = e^{-1}$$

$$f'(c) = -\sin c \cdot e^{\cos c} = 0$$

$$f''(c) = -\cos c \cdot e^{\cos c} = e^{-1}$$

$$\text{or } f(x) \approx e^{-1} + \frac{1}{2!} \cdot e^{-1} (x-\pi)^2$$

$$\boxed{f(x) \approx \frac{1}{e} + \frac{1}{2e} (x-\pi)^2} \quad \square$$

Taylor polynomial
for degree 2 around $x=\pi$

_____ x _____ x _____ x _____

2. Let a sequence x_n be defined inductively by $x_{n+1} = F(x_n)$. Suppose that $x_n \rightarrow x$ as $n \rightarrow \infty$ and $F'(x) = 0$. Show that

$$x_{n+2} - x_{n+1} = \mathcal{O}(x_{n+1} - x_n)$$

Hint: Use the Mean-Value Theorem and assume that F is a continuously differentiable function.

TPT $x_{n+2} - x_{n+1} = \mathcal{O}(x_{n+1} - x_n)$

LHS = $x_{n+2} - x_{n+1}$

(1a)

or LHS = $F(x_{n+1}) - F(x_n)$

(1b)



Using MVT on x_n, x_{n+1} :

$$F(x_{n+1}) - F(x_n) = F'(\xi) \cdot (x_{n+1} - x_n)$$

where $\xi \in (x_n, x_{n+1})$
 $\downarrow$ between

or $x_{n+2} - x_{n+1} = F'(\xi) (x_{n+1} - x_n)$

$\xi \in (x_n, x_{n+1})$ (2)

Now given F is continuously differentiable, if we know that $F'(y) = L$ for some y and a finite L , we can safely argue that the value of F' cannot change abruptly around the neighbourhood of y and will be bounded as well.

In ②, as $n \rightarrow \infty$,

$$u_{n+1} \rightarrow n, u_n \rightarrow n$$

(!! does not mean $u_{n+1} = n$ or $u_n = n$ or even $u_{n+1} = u_n$),

$$\xi \rightarrow n.$$

$\therefore F'(n) = 0$ ^{finite} and ξ is in the neighbourhood of n , $F'(\xi) \leq M$, a finite bound.

$\therefore$ ② may be written as

$$u_{n+2} - u_{n+1} \leq M \cdot (u_{n+1} - u_n)$$

in fact $n \rightarrow 0$ as $n \rightarrow \infty$.

Thus, $\frac{u_{n+2} - u_{n+1}}{u_{n+1} - u_n} \rightarrow 0$ as $n \rightarrow \infty$

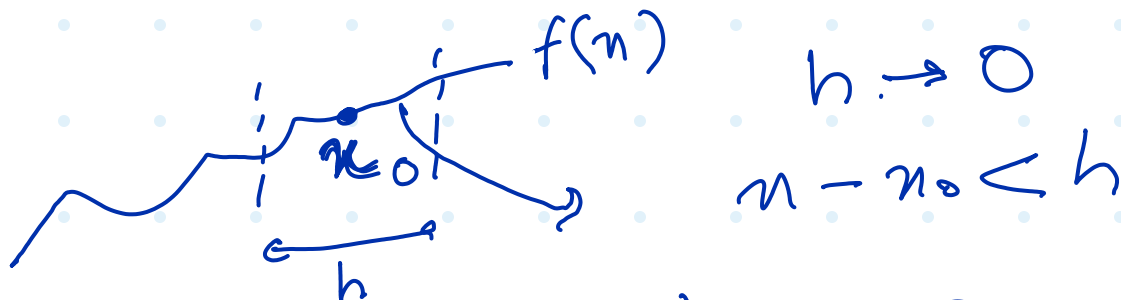
$$\therefore \underbrace{u_{n+2} - u_{n+1}}_{\text{order of } u_{n+2} - u_{n+1}} = \underbrace{O(u_{n+1} - u_n)}_{\substack{\text{strictly} \\ \text{smaller} \\ \text{than} \\ \text{negligible} \\ \text{compared} \\ \text{to}}} \underbrace{\quad}_{\substack{\text{order of} \\ u_{n+1} - u_n}} \quad \square$$

X

X

X

3. Prove that every sufficiently smooth function can be approximated on an interval of length h by a polynomial of degree n with an error that is $\mathcal{O}(h^{n+1})$ as $h \rightarrow 0$.



We can attempt to use Taylor's theorem for $f^{(n)}$ of order n around any arbitrary $n = n_0$.

$$f(x) = \sum_{k=0}^n \frac{1}{k!} f^{(k)}(x_0) \cdot (x - x_0)^k + \frac{1}{(n+1)!} f^{(n+1)}(\xi) \cdot (x - x_0)^{n+1}$$

$x = n$
 $c = x_0$

$\xi \in (x_0, n)$

As we're given that f is "sufficiently smooth", we can assume that $f', f'', \dots, f^{(n)}$ and $f^{(n+1)}$ are bounded in the neighbourhood of x_0 . ①

i.e. $f'(x_0) \leq M_1, f''(x_0) \leq M_2, \dots, f^{(n)}(x_0) \leq M_n$
 for finite $M_i, i=1$ to n and
 where $|n - x_0| < h$.

Also $\because \xi \in (x_0, x)$

ξ is also in neighbourhood of x_0

and thus $f^{(n+1)}(\xi) \leq M$.

Thus LHS of (1) may be upper
 bounded by switching all
 derivative and difference
 terms with their upper bounds.

$$f(x) \leq \sum_{k=0}^n \frac{1}{k!} M_i \cdot h^k + \frac{1}{(n+1)!} M \cdot h^{n+1}$$

(2)

Thus the non-polynomial
 error term is $O(h^{n+1})$
 as $h \rightarrow 0$.

□

22. Prove that if $0 < \theta < 1$, then $(1 + a\theta^n)/(1 + a\theta^{n-1})$ converges to 1 linearly.

$$u_n = \frac{1 + a\theta^n}{1 + a\theta^{n-1}} \quad \theta \in (0, 1)$$

$$\text{Let } L = \lim_{n \rightarrow \infty} u_n$$

$$\text{or } L = \frac{1 + a \cdot 0}{1 + a \cdot 0} = 1 \quad \text{as } \theta^n \rightarrow 0 \quad \because \theta \in (0, 1)$$

$$\text{or } \boxed{L = 1}$$

$$e_n = |1 - u_n|$$

$$\text{or } e_n = \left| 1 - \frac{1 + a\theta^n}{1 + a\theta^{n-1}} \right|$$

$$\text{or } e_n = \left| \frac{1 + a\theta^{n-1} - 1 - a\theta^n}{1 + a\theta^{n-1}} \right|$$

$$\text{or } e_n = \left| \frac{\overbrace{a\theta^{n-1}}^{>0} (\overbrace{(1-\theta)}^{>0})}{\underbrace{1 + a\theta^{n-1}}_{>0}} \right|$$

$$a \quad r_n = \frac{a\theta^{n-1}(1-\theta)}{1+a\theta^{n-1}}$$

$$r_{n+1} = \frac{a\theta^n(1-\theta)}{1+a\theta^n}$$

for linear convergence,

$$\lim_{n \rightarrow \infty} \frac{r_{n+1}}{r_n} = c \quad \text{s.t.} \quad c \in (0,1)$$

$$a \quad \lim_{n \rightarrow \infty} \frac{\cancel{a\theta^{n-1}}(1-\cancel{\theta}) / (1+a\theta^n)}{\cancel{a\theta^{n-1}}(1-\cancel{\theta})(1+a\theta^{n-1})}$$

$$a \quad \lim_{n \rightarrow \infty} \frac{\theta(1+a\theta^{n-1})}{(1+a\theta^n)}$$

$$a \quad \lim_{n \rightarrow \infty} \frac{\theta(1+a \cdot 0)}{(1+a \cdot 0)} = \theta \quad (\text{a constant}) \in (0,1)$$

$$\therefore \left| \frac{r_{n+1}}{r_n} \right| \rightarrow \theta \text{ as } n \rightarrow \infty, \\ \therefore \quad \text{converges linearly} \quad \square$$